Cumulative Incidence Curve Regression

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2026-02-22

Purpose

This vignette illustrates regression of cumulative incidence probabilities via pseudo-values, using **CICs** for the CIC and pseudo-value calculations.

```
suppressPackageStartupMessages({
  library(CICs)
  library(dplyr)
  library(ggplot2)
})
```

Pseudo-value regression

Simulate data

We generate two-arm competing risks data using **CICs** `GenTwoSampleData`, which calls `GenData` for each arm. Arm 1 has a lower event rate (0.35) than arm 0 (0.50), with the same death and censoring rates in both arms.

```
set.seed(101)
df <- GenTwoSampleData(
  n1 = 200,
  n0 = 200,
  event_rate1 = 0.35,
  event_rate0 = 0.50,
  death_rate1 = 0.25,
  death_rate0 = 0.25,
  censor_rate1 = 0.25,
  censor_rate0 = 0.25,
  tau = 10
)

df %>%
  dplyr::group_by(arm) %>%
  dplyr::summarise(
    n = dplyr::n(),
    n_censor = sum(status == 0),
    n_event = sum(status == 1),
    n_death = sum(status == 2),
    .groups = "drop"
  )
#> # A tibble: 2 x 5
#>     arm     n n_censor n_event n_death
#>   <dbl> <int>    <int>   <int>   <int>
```

```
#> 1      0   200        42      103       55
#> 2      1   200        51       78       71
```

Pseudo-values

The pseudo-value for subject i at time τ is

$$\hat{\theta}_i(\tau) = n \cdot \hat{\theta}(\tau) - (n-1) \cdot \hat{\theta}_{(-i)}(\tau),$$

where $\hat{\theta}(\tau)$ is the CIC at τ from the full sample and $\hat{\theta}_{(-i)}(\tau)$ is the CIC at τ from the sample with subject i removed. We use **CICs** `CalcCIC` for both.

```r
# CIC at time tau (step function: value at last time <= tau).
cic_at_tau <- function(time, status, tau) {
  fit <- CalcCIC(status = status, time = time)
  idx <- which(fit$time <= tau)
  if (length(idx) == 0) return(0)
  fit$cic_event[max(idx)]
}

# Pseudo-values at tau via jackknife (using CICs::CalcCIC).
gen_pseudo_cic <- function(data, tau) {
  n <- nrow(data)
  time <- data$time
  status <- data$status
  theta_full <- cic_at_tau(time, status, tau)
  theta_loo <- vapply(seq_len(n), function(i) {
    cic_at_tau(time[-i], status[-i], tau)
  }, numeric(1))
  data$pseudo <- n * theta_full - (n - 1) * theta_loo
  data
}
```

Fit pseudo-value regression

Pseudo-values at $\tau = 2$ are modeled with linear regression on treatment arm; the coefficient is the risk difference at τ.

```r
tau <- 2
df <- gen_pseudo_cic(df, tau = tau)
fit <- stats::lm(pseudo ~ arm, data = df)
summary(fit)
#>
#> Call:
#> stats::lm(formula = pseudo ~ arm, data = df)
#>
#> Residuals:
#>     Min      1Q  Median      3Q     Max
#> -0.6334 -0.4807 -0.1863  0.5727  1.0582
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  0.51236    0.03806  13.460  < 2e-16 ***
#> arm         -0.15265    0.05383  -2.836  0.00481 **
#> ---
```

```
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 0.5383 on 398 degrees of freedom
#> Multiple R-squared:  0.0198, Adjusted R-squared:  0.01734
#> F-statistic: 8.041 on 1 and 398 DF,  p-value: 0.004806
```

The negative coefficient on arm indicates that treatment (arm 1) is associated with a lower cumulative incidence of the event at $\tau = 2$, consistent with the lower event rate in arm 1.

Ground truth (no censoring)

With no censoring, a larger sample gives a precise estimate of the risk difference at τ. We use `GenTwoSampleData` with `censor_rate` 0 and a truncation time so that all events are observed.

```
set.seed(101)
df_gt <- GenTwoSampleData(
  n1 = 500,
  n0 = 500,
  event_rate1 = 0.35,
  event_rate0 = 0.50,
  death_rate1 = 0.25,
  death_rate0 = 0.25,
  censor_rate1 = 0,
  censor_rate0 = 0,
  tau = 10
)
df_gt <- gen_pseudo_cic(df_gt, tau = 2)
fit_gt <- stats::lm(pseudo ~ arm, data = df_gt)
summary(fit_gt)
#>
#> Call:
#> stats::lm(formula = pseudo ~ arm, data = df_gt)
#>
#> Residuals:
#>    Min     1Q Median     3Q    Max
#> -0.514 -0.406 -0.406  0.486  0.594
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  0.51400    0.02218  23.174  < 2e-16 ***
#> arm         -0.10800    0.03137  -3.443 0.000599 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 0.496 on 998 degrees of freedom
#> Multiple R-squared:  0.01174,    Adjusted R-squared:  0.01075
#> F-statistic: 11.85 on 1 and 998 DF,  p-value: 0.000599
```